



RoClean™

Contact Web Cleaning

RoClean is available as a single-sided cleaning system, with double-sided cleaning configuration available as an option. The RoClean is suitable for high-performance contamination removal from low-tension web applications.

The system uses a combination of negative and positive airflows and a rotating bristle brush that breaks the boundary layer from the material surface and removes loose contaminants or contaminants that are bound by electrostatic charges; leading to a clean surface. The airflows of the RoClean are expertly controlled to optimise the cleaning performance.

RoClean is supplied with the latest Hyperion™ 924IPS static control bars, to maximise the cleaning performance.

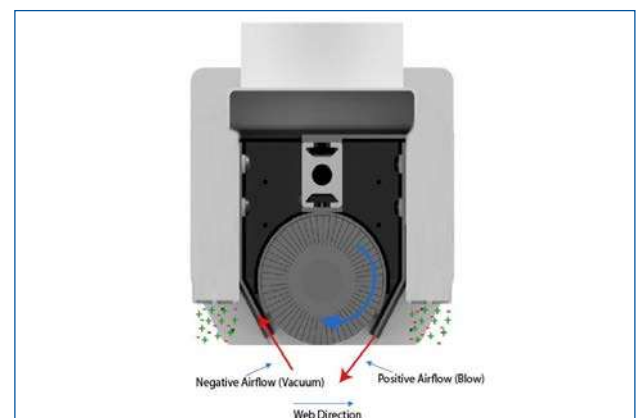
Installation



The RoClean head unit is installed over a guide roller. It may be positioned at any angle to best suit the installation, whilst still delivering uncompromised surface cleaning. The RoClean overcomes tension issues, as it is designed to be positioned over a roller, where the tension in the web is naturally at it's highest.

How RoClean Works

- The web passes through a cloud of ionisation on entry to the RoClean head, which neutralises any static charges on the web.
- The contact rotary brush rotates in the opposite direction of the web, therefore disturbing the boundary layer of the web and loosening the contaminants which are then extracted into the vacuum chamber. Before the web exits the RoClean, a positive air wave passes over the material, ensuring that any contamination on the rotary brush is also moved to the vacuum chamber. This ensures that when the web exits from the RoClean it is as clean as possible. Contamination is then trapped in the filter located in the Air Handling Unit to prevent airborne contamination.
- As the web exits the RoClean head, it passes through a second cloud of ionisation to prevent re-contamination of the web.



RoClean Web Cleaner (cross section of airflow and brush direction)

Applications

Meech developed RoClean to target the battery film production market where high levels of cleaning efficiency are crucial for the quality of the end product. That said, RoClean is suitable for a range of industrial applications.

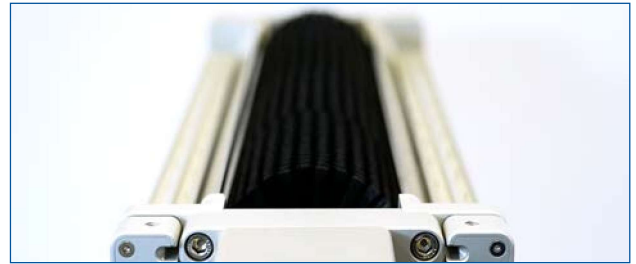
RoClean can be used within a wide variety of applications and with the range of bristle material available, it is suitable for use with all web materials.

Typical applications include:

- Battery - Aluminium foil
- Printing: Flexo, Digital, Gravure and Ink-Jet
- Coating
- Laminating
- Food and medical packaging
- Specialty film processing
- Solar panel substrates
- Blister packs

Design Features

The RoClean is a robust, yet compact system. Based on our unrivalled experience and extensive development program, we have included several unique features in the system to ensure excellent performance:



RoClean showing brush

Feature	Benefit
Hyperion™ Pulsed DC Ionising technology offered as standard (integrated)	The latest DC static control technology neutralises the web surface as it enters and exits the RoClean. This reduces the risk of re-contaminating the web.
Adjustable rotary brush height	The height of the rotating brush roller is adjustable up to 5mm, assisting the initial setup and ensuring RoClean is flexible to work with various web thicknesses.
Long rotary brush operating life	Ensures minimum disruption of the production line and maintaining high levels of efficiency.
Easy-change of rotary brushes	The ability to quickly change the rotary brush on RoClean is available from both ends of the unit, ensuring easy access and keeping the idle time to minimum.
Unique feature to clean the rotating brush	Ensures optimum cleaning results
Roller speeds adjustable from the AHUv3	The roller speed is adjusted and maintained by the touchscreen available on the AHUv3*
CleanRoom ISO 7 Class compatible	Specially designed to work in clean and dry room environments**
No compressed air required	Keeps operating cost to a minimum

* Only available when the control panel is included with the system

** Definition of drying room conditions: Purity class according to ISO 14644-1: ISO 7, Ambient Temperature +20 °C ± 2 °C, Dew Point -55 °C ± 2 °C, Room Humidity: <1%

Scope of Supply

Feature	Benefit
Hyperion™ Pulsed DC Ionising technology offered as standard (integrated)	The latest DC static control technology neutralises the web surface as it enters and exits the RoClean. This reduces the risk of re contaminating the web.
Control Panel (optional)	Connects the Air Handling Unit with the RoClean for control via the AHU.
Brushes	Standard brushes made from a low water absorption material that will not risk damage to the material cleaned. Easily removable from both ends of the RoClean. Other materials available on request.
Air Handling Unit (AHUv3 & AHUv3.5)	Provides the positive and negative airflow for the RoClean and the filtration. Additionally provides control over the RoClean. AHUv3.5 is designed to handle toxic contamination.



TakClean™

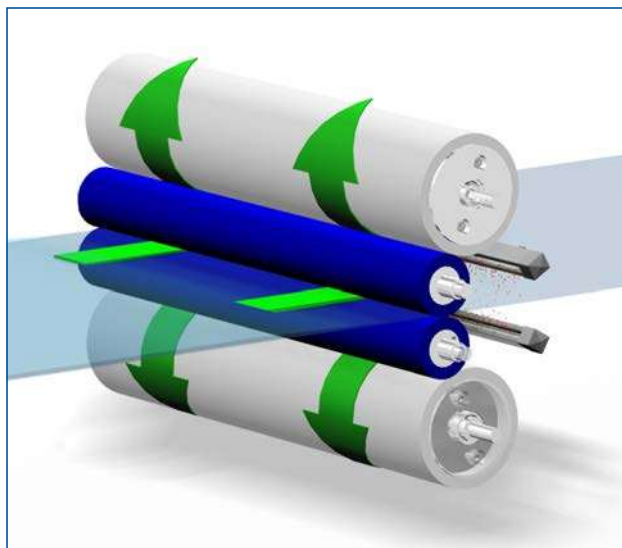
Contact Web Cleaning

The Meech TakClean contact web cleaning system is designed to remove dry, unbonded contamination from a moving web.

TakClean incorporates the specially formulated TransTak elastomer contact cleaning rollers and perforated adhesive rolls which combined, make TakClean the ultimate elastomer-based web cleaning system.

Meech has designed TakClean to provide the following user advantages:

- Excellent contamination removal – removing particles as small as 0.5 micron.
- Continuous cleaning – no need to stop the web.
- Fast installation – very low downtime for installation.
- Static control – web will be contamination free and neutralised on exit.



How TakClean Works

The unrivalled cleaning performance of TakClean is the result of 3 critical processes:

- The TransTak elastomer cleaning rollers make contact with the full width of the web with even pressure. Dry, unbonded contamination is lifted from the web surface.
- Pneumatics hold the adhesive roller in contact with the TransTak elastomer roller. Contamination is transferred from the surface of the TransTak elastomer roller to the adhesive roller where it is trapped.
- Any static charge present on the web is neutralised on exit from the TakClean; minimising the risk of statically attracted recontamination.

TransTak Elastomer Rollers

The specially formulated TransTak elastomer rollers are available in a variety of Shore A ratings (35 standard). TransTak rollers include cured silicone, but crucially are proven to have no 'free chemistry' and therefore do not leave any trace on the surface being cleaned.



Adhesive Rolls

The quality and performance of the adhesive rolls in any elastomer based web cleaning machine is vital to achieve the best levels of contamination removal.

Meech adhesive rolls are manufactured from a moisture stabilised paper base with a release coating to ensure easy unwinding. The paper is coated with an adhesive which is specially formulated to work effectively with TransTak elastomer rollers. Meech adhesive rollers have a high adhesive mass, the result of which is longer and more effective cleaning.

Meech adhesive rolls are a continuous roll of material with perforations in set positions along the length. The positioning of the perforations gives 70 easily removable (tear off) sections. The major advantages of Meech perforated rolls are as follows:

- A knife is not required to cut the material, leading to improved operator safety and no risk of damage to the TakClean system.
- No wasted adhesive material: the operator only removes a pre-specified length of adhesive roll material.

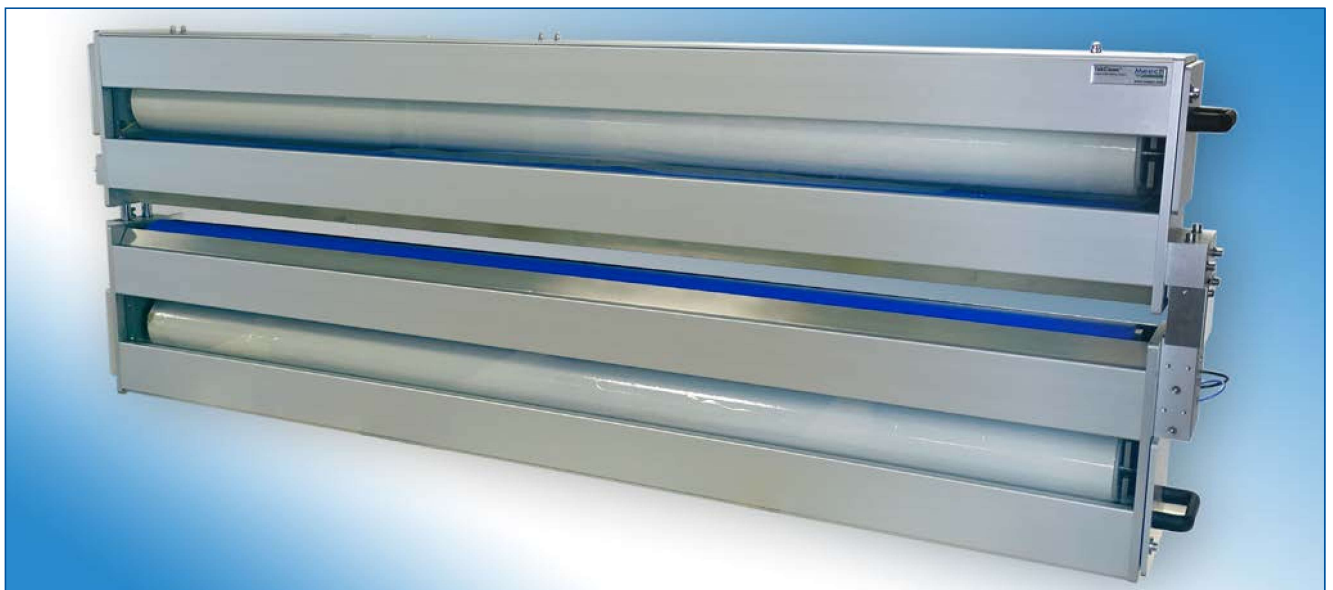
Adhesive Roll Specification

Material	Paper
Coatweight	High; accurately controlled
Core size	76mm / 3" Ø
Core material	High impact polystyrene
Material length per roll	22.4mtrs / 73ft
Perforations	Roll is perforated into 70 removable sections
Material widths	Any width from 100-1500mm

Design Features

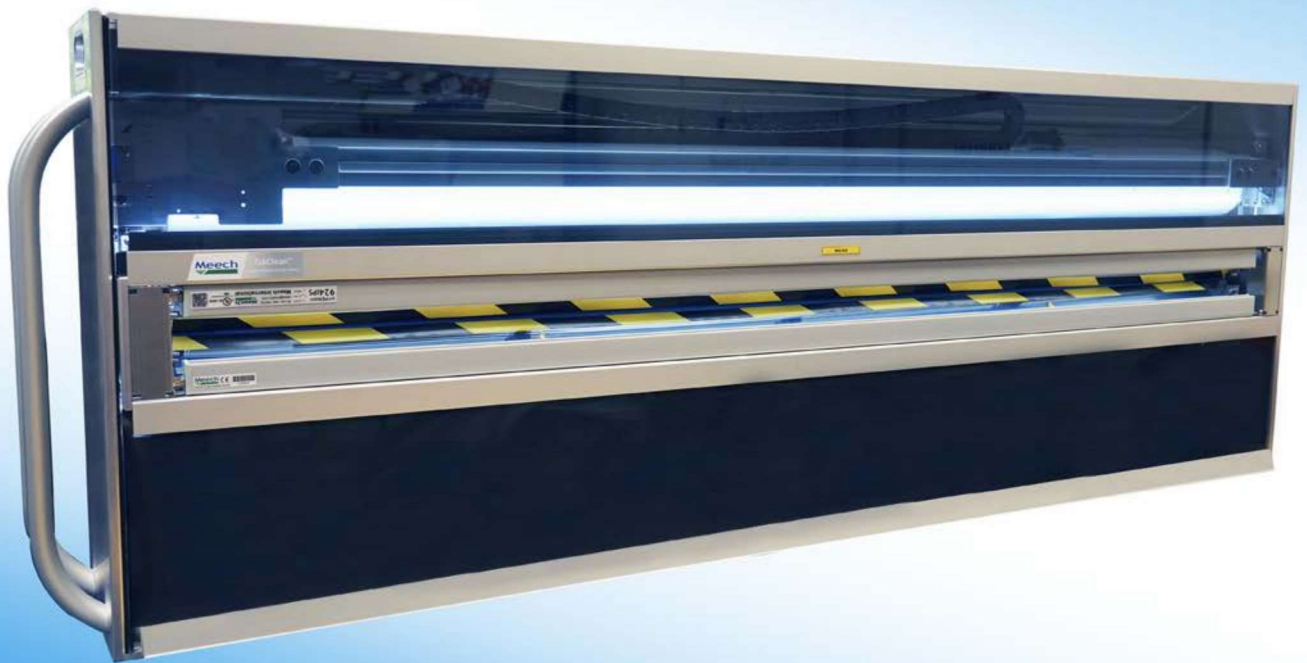
TakClean is constructed from hard anodised aluminium extrusions and machined hard anodised components to provide a highly rigid design with exceptional build quality. In particular TakClean incorporates the following main features:

Feature	Benefit
Clam shell opening (top half) for narrow web	Provides the operator with access to clean the TransTak elastomer rollers and makes webbing-up very easy.
Side pull-out adhesive rolls	Minimises the size of the machine and provides the operator with easy access to remove a section of the perforated adhesive roll.
Cantilever or baseplate mounting	Flexibility of the installation position. TakClean can be installed to suit all web orientations.
Integrated static control	Meech active static control equipment neutralises the web on exit of the TakClean to prevent statically attracted recontamination.
Fast change adhesive rolls	Reduces maintenance time required.



"Working with Meech is an obvious choice for Edale. It enables us to offer integrated solutions to suit our customers' exacting needs."

Design Office Manager, **Edale**



TakClean Ultra™

Contact Web Cleaning

The Meech TakClean Ultra is a re-design of the hugely successful Meech TakClean system.

Manufactured specifically for medium and wide webs between 650 and 1,450mm, the TakClean Ultra was designed following extensive R&D. Built to the highest quality standard, the TakClean Ultra delivers advanced tacky roller cleaning at wider widths.



TransTak Elastomer Rollers

The specially formulated TransTak elastomer rollers are available in a variety of Shore A ratings (35 standard). TransTak rollers include cured silicone, but crucially are proven to have no 'free chemistry' and therefore do not leave any trace on the surface being cleaned.



How TakClean Ultra Works

The TakClean Ultra includes a number of new features not seen in the original design, but at its core the TakClean Ultra is still developed from the same tried and tested technology of the TakClean.

- Pneumatics are used to hold the TransTak Elastomer cleaning rollers in place. This allows them to offer an even and consistent level of pressure across the full width of the web. Dry, unbonded contamination is then lifted from the web surface.

- Pneumatics are also used to hold the adhesive roller against the TransTak Elastomer roller. Contamination is transferred from the surface of the TransTak roller to the adhesive roller, where it is trapped.

- Any static present on the web is neutralised on exit from the TakClean Ultra by 2 x Hyperion 924IPS bars. This minimises the risk of recontaminating the web.

- TakClean Ultra can be webbed up just by disengaging the pneumatic elastomer rollers. This makes the TakClean Ultra more user-friendly and easier to maintain, without the need for tools.



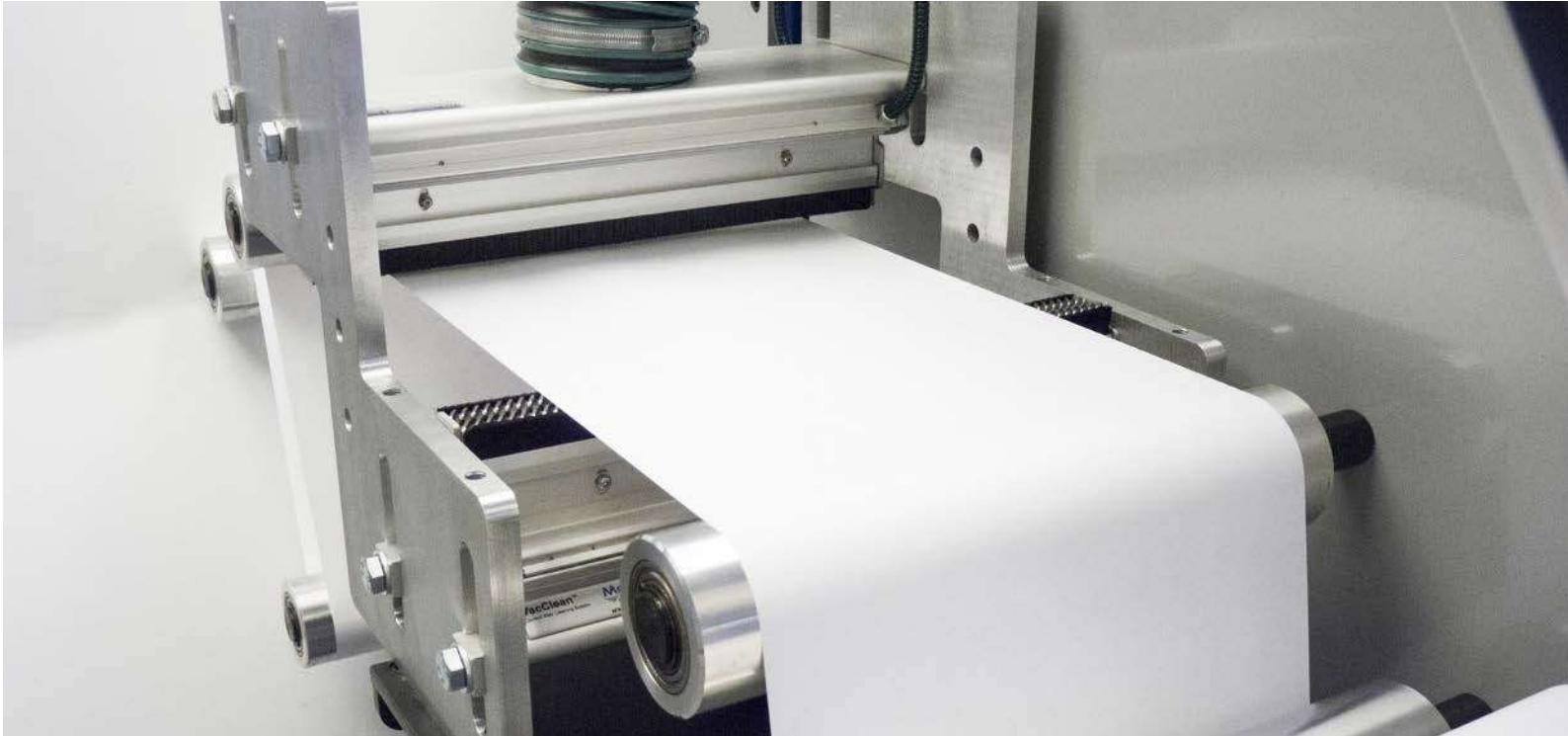
Design Features

TakClean Ultra is constructed from hard anodised aluminium extrusions and machined hard anodised components to provide a highly rigid design, which is crucial when dealing with machinery designed for wide web applications. It also boasts exceptional build quality.

TakClean Ultra has been meticulously designed and engineered to incorporate a number of features within its design, which makes it ideal for a range of different applications using medium and wide webs:



Feature	Benefit
Compact Design	Every component is self contained within the TakClean Ultra, so there is no additional space requirements. This means the TakClean Ultra can fit into areas that require a small footprint.
Hyperion™ Pulsed DC Ionising technology offered as standard	The latest DC static control technology neutralises the web surface as it exits the TakClean Ultra. This reduces the risk of recontaminating the web. Automatic ionisation means the static control will only be switched on when the elastomer roller is engaged. This will increase the life of the static control equipment.
Independent TransTak Elastomer and adhesive rollers	Independent rollers mean that you can disengage the adhesive roller whilst the Elastomer roller is still engaged, allowing for efficient and uninterrupted cleaning. Changing the adhesive roller can be done easily and quickly, reducing downtime and making webbing up easy and contact free.
Automatically adapts to web thickness	Provides flexibility and allows different materials to run on the same production line.
Automatic mode with manual override	Allows the TakClean Ultra to be controlled either manually, via the direct TakClean Ultra controls, or automatically via independent external controls. Both the elastomer and adhesive rolls provide remote signals.
Standard and reverse direction of web travel	The smart design of the TakClean Ultra allows the 924IPS bars to be easily mounted on either side of the unit. This means the direction of the web can be reversed at anytime, providing maximum user flexibility.
Double-sided and single-sided cleaning	To suit a range of applications.
Elastomer rollers separate by 25.4 mm / 1 inch	Making webbing up easy.
Safety Guarding	Provides protection against accidents, whilst also protecting the adhesive rolls from external contamination.
Rollers can be changed without the need for any tools	Allows for quick and easy set-up and maintenance.
Inspection LED	LED lights built into the TakClean Ultra show any contamination left on the web. This allows the user to identify issues immediately.
Hard-wearing	The design of the TakClean Ultra means there is very little wear on parts. This will extend the life of the System.
Self-aligning rollers	Allows easy set up of the pneumatics.



VacClean™

Contact Web Cleaning

The Meech VacClean contact web cleaning system is the ultimate system for the removal of high levels of contamination.

VacClean systems incorporate an enhanced contact cleaning manifold with either an Air Handling Unit (AHUv3) or vacuum fan unit, depending on the installation environment and web width.

Meech engineers created the VacClean manifold profile with the following design characteristics:

- Highly polished stainless steel faceplates which create turbulence to break the boundary layer.
- The delta profile, which is proven to have excellent airflow characteristics, ensuring that removed contamination is efficiently transferred to the system filter.
- A compact web contact point that means VacClean can be installed in areas with restricted space.
- Twin ionising bars, which are angled for an increased ionisation footprint, to neutralise static charges on the web on entry and exit to the manifold.
- Brushes that aid the break-up of the boundary layer without contacting the web surface.

For a combined manifold length up to 1,650mm/65" the VacClean is supplied with an AHUv3, which provides the airflow and filtration for the system.

A combined manifold length of 1,650mm/65" or greater requires the VacClean 3-phase centrifugal vacuum fan unit, which includes an outlet silencer and 5 micron contamination collection bag (upgradable to 1 micron).

If preferred, the vacuum fan unit is also available as an option for installations with a combined manifold length of under 1,650mm/65".

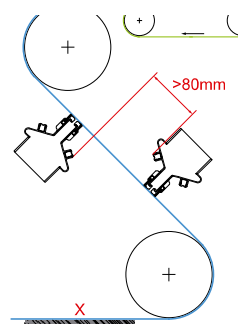
Applications

VacClean can be used in a wide variety of applications where contact cleaning is required. Typical applications include:

- | | |
|----------------------|-------------------------------|
| ■ Printing | ■ Corrugating |
| ■ Labelling | ■ Food / medical packaging |
| ■ Laminating | ■ Paper making / converting |
| ■ Coating | ■ Wood / furniture / flooring |
| ■ Slitting & winding | ■ Automotive |

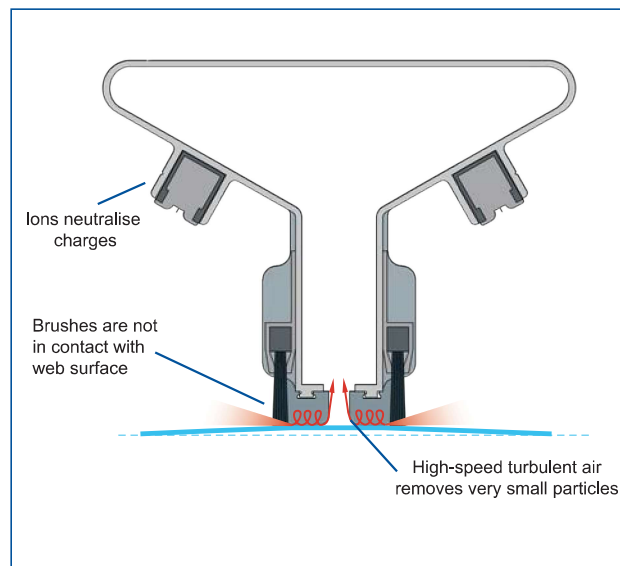
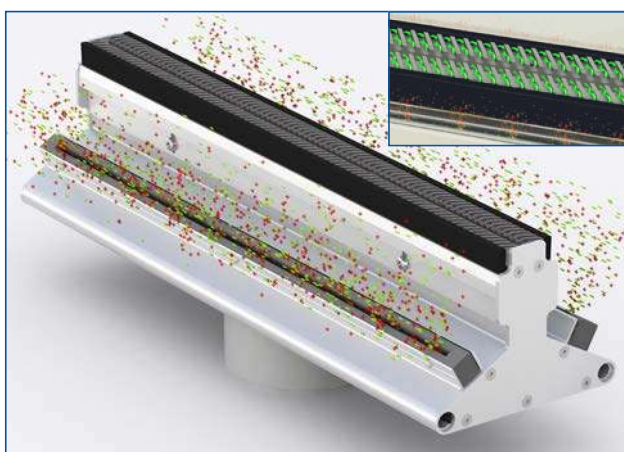
Installation

The VacClean manifold/s can be installed on a web travelling at any angle without jeopardising cleaning performance. The manifold/s should be installed between rollers (typically no closer than 100mm/4") in an area where the web remains under tension. If the system is double sided, the manifolds should be spaced by a minimum of 80mm/3". The web must be in free air, i.e. away from rollers, conveyors or the machine bed.



How VacClean Works

- The web passes through a cloud of ionisation which neutralises any static charges present.
- The brushes just contact the surface of the material and break up the boundary layer of air, enabling effective removal of the smallest particles.
- The web then fully contacts two highly polished vacuum faceplates that give complete coverage of the web, creating highly turbulent airflows and concentrating the vacuum to remove contaminants with a very high degree of efficiency.
- As the web leaves the VacClean manifold, it passes through a second cloud of ionisation to prevent re-contamination of the web.



Cross section of VacClean showing operation

Filtration

Excellent filtration is paramount in any web cleaning system. The VacClean system has the following standard filtration levels:

- System including AHUV3 (Compact or Standard) = F8 grade bag filter. Can be upgraded to a H14 grade HEPA filter.
- System including vacuum fan = 5 micron filter bag. Can be upgraded to a 1 micron filter bag.

Design Features

The VacClean manifold is constructed from anodised extruded aluminium, making it highly rigid and suitable for single manifolds to span web widths up to 3980mm, 156". Manifolds can be joined to span wider webs.

Feature	Benefit
Compact contact point	Cleaning in areas of restricted space can be achieved.
Integrated static control	Neutralises static charges to optimise cleaning performance and prevent re-contamination.
Polished stainless steel faceplates	Creates turbulence to break boundary layer and remove contamination.
No consumable items	Low ongoing costs.
Automatic vacuum pressure control	Reduces the frequency of required operator adjustments. Maintains optimum cleaning performance automatically.

Options

VacClean systems can be manufactured to include a range of options including:

- Vacuum fan pre filters to allow fan discharge air to be exhausted to atmosphere.
- Cyclonic dust collection.
- Acoustic enclosures to provide very low vacuum fan noise levels.

"Since installation of the VacClean system we have seen a significant reduction in slitting dust on our final product."

Technical Manager, **Gascogne Paper**